Cisco Packet Tracer

Level 1 – Basic Network Setup & Connectivity

Overview

Cisco Packet Tracer is a free network simulation tool developed by Cisco. It lets you design, build, and test computer networks visually without needing any real physical hardware. It is one of the most widely used tools for studying networking and is essential for anyone working toward a Network or CCNA certification.

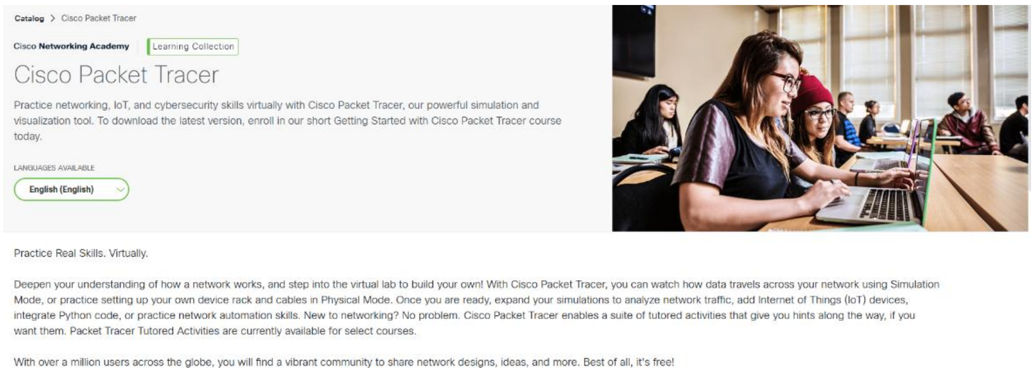
This first section covers getting Packet Tracer installed, understanding the interface, and building a basic two-PC network connected through a switch – including IP configuration and connectivity testing using ping.

What will you learn:

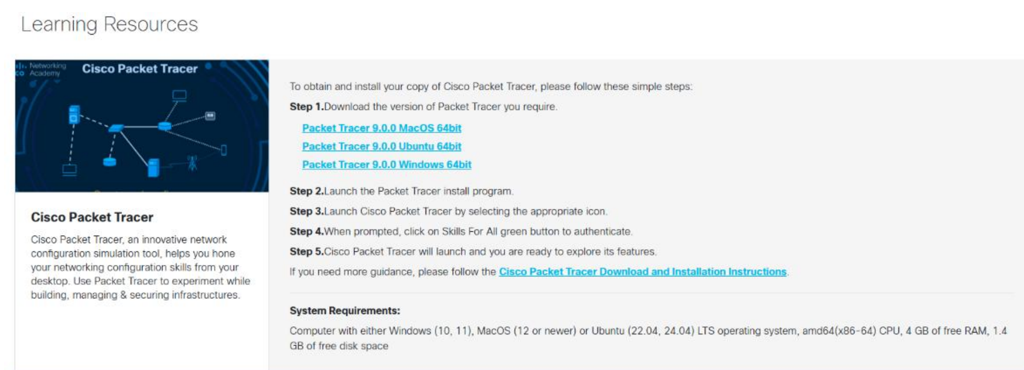
- What is Cisco Packet Tracer and how to download it
- How to navigate the Packet Tracer interface
- How to build a basic network topology with a switch and two PCs
- How to assign static IP addresses to devices
- How to test connectivity between devices using ping
- How to use Simulation Modet to visually see network traffic

Part 1 – Downloading Cisco Packet Tracer

Packet Tracer is completely free and available through Cisco's NetAcad platform.



1. Go to **netacad.com** and log in or create a free account.
2. From the top menu, click **Courses** and search for **Packet Tracer**.
3. Click on **Cisco Packet Tracer (8hrs)**, then click **Getting Started with Cisco Packet Tracer.**
4. Enroll on the course, it is free. Navigate to module **1.0.3** where you will find the download link.



5. Scroll down to find the download steps. Choose the version appropriate for your operating system and download it.
6. Once downloaded, run the installer and follow the standard installation steps. Click **Finish** and launch Packet Tracer.
7. When prompted, log in using your **Cisco NetAcad account.** The app requires an active login to use.

Note: If you already have a NetAcad account from your CISCO certifications, use the same login. No need to create a new one.

Part 2 – Understanding the Interface

Before building anything, it helps to know where everything is in the Packet Tracer workspace.

Device Menu (Bottom Panel)

- **Main Categories (left side):** Network Devices, End devices, Components Connections, Miscellaneous, and Multiuser Connection
- **Sub Categories**: Routers, Switches, Hubs, Wireless Devies, Security, and WAN Emulation
- **Device List (middle)**: Shows available devices based on the selected category. Also has a search bar for quick access

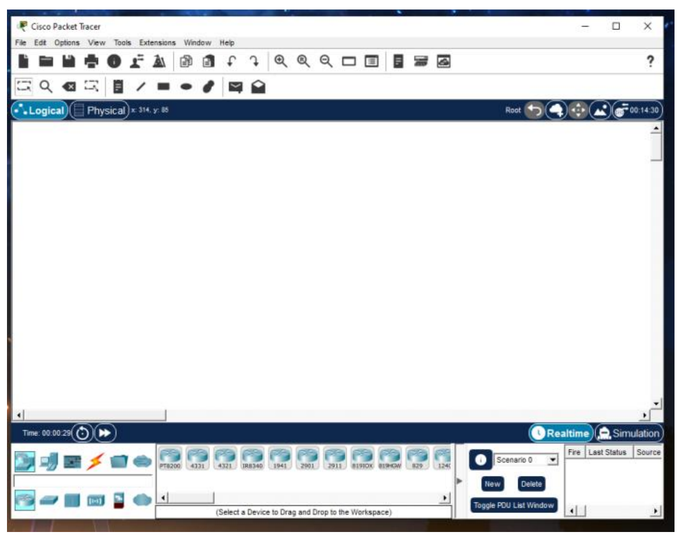
Workspace

The large blank area in the center is the workspace where you drag and drop devices to build your network topology.

Logical View vs. Physical View

- **Logical View –** the default view where you spend most of your time. Shows how devices connect to each other logically

- **Physical View –** shows a realistic simulation of what the network would look like physically, like a city with buildings, offices, and wiring closets. Good for visualizing real world scale but most work happens in Logical View



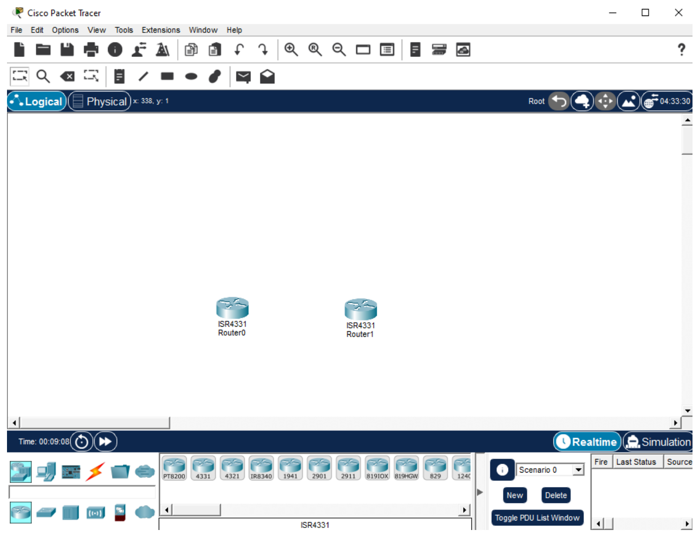
Toolbar (Top Left)

- **Select Tool –** select and moved devices
- **Inspect Tool –** quickly view device information like routing tables
- **Delete Tool –** remove devices from the workspace
- **Place Note Tool –** add text labels to the diagram
- **PDU Tools –** add test data to the network for quick connectivity checks

Part 3 – Building a basic topology

The goal here is to build a simple network with two PCs connected to one switch and verify if they can communicate with each other.

Add Devices to the Workspace

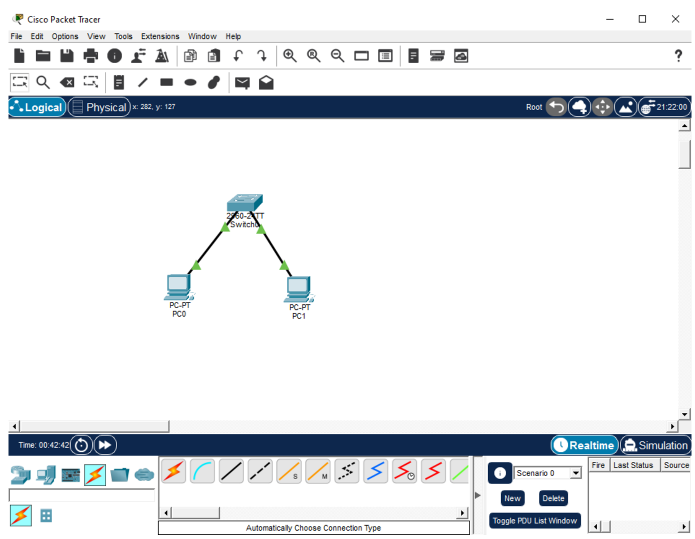


8. In the bottom left, click **Network Devices,** then select **Switches**.

9. Drag a switch onto the workspace. Any switch model works. If unsure, just use the highlighted default one.

10. Click **End Devices,** then select **PC**. Drag two PCs into the workspace. Label them **PC0 and PC1**.

Connect the Devices



11. Click the **Connections** icon on the bottom left and select **Copper Straight-Through** cable.

12. Click on **PC0,** select **FastEthernet0,** then click on the **Switch** and select **FastEthernet0/1.**

13. Repeat for **PC1.** Connect it to **FastEthernet0/2** on the switch.

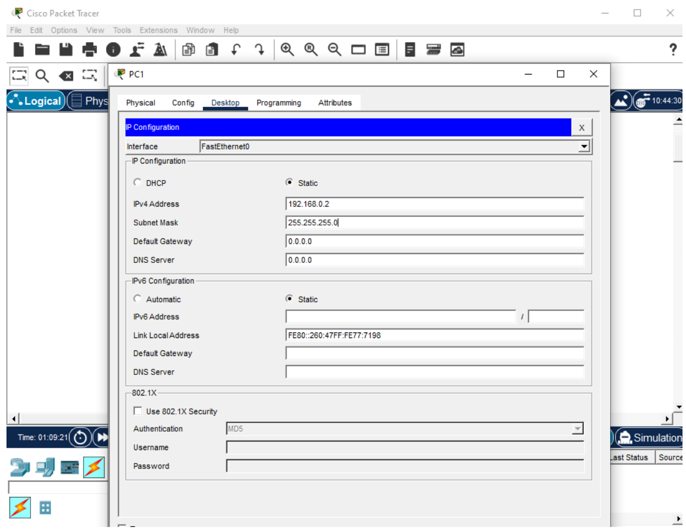
Best Practice: If you are unsure which cable type to use, click the lightning bolt icon (Automatically Choose Connection Type). Packet Tracer will automatically pick the right cable for the selected devices.

Part 4 – Configuring IP Addresses

Each device needs a unique IP address to communicate on the network. Here we assign them manually using static IPs.

Configure PC0

14. Click on **PC0** to open its properties window.
15. Go to the **Desktop** tab and click **IP Configuration.**
16. Select **Static** and enter the following:
    - **IPv4 Address:** 192.168.0.1
    - **Subnet Mask: 255.255.255.0** (this autofills)
17. Close the window.



Configure PC1

18. Click on **PC1** and go to **Desktop > IP Configuration**
19. Select **Static** and enter:
    - **IPv4 Address:** 192.168.0.2
    - **Subnet Mask:** 255.255.255.0
20. Close the window.

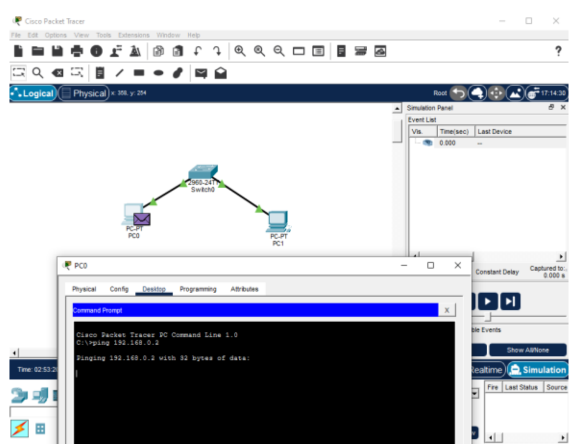
Note: Both PCs must be on the same network. Notice that **192.168.0.1** and **102.168.0.2** are both in the 192.168.0.0/24 network. This is what allows them to communicate directly without needing a router.

Part 5 – Testing Connectivity with Ping

With Ips configured, the next step is to verify the two PCs can communicate with each other.

21. Click on **PC0** and go to **Desktop > Command Prompt**

22. Type the following command and press enter:
    ping 192.168.0.2



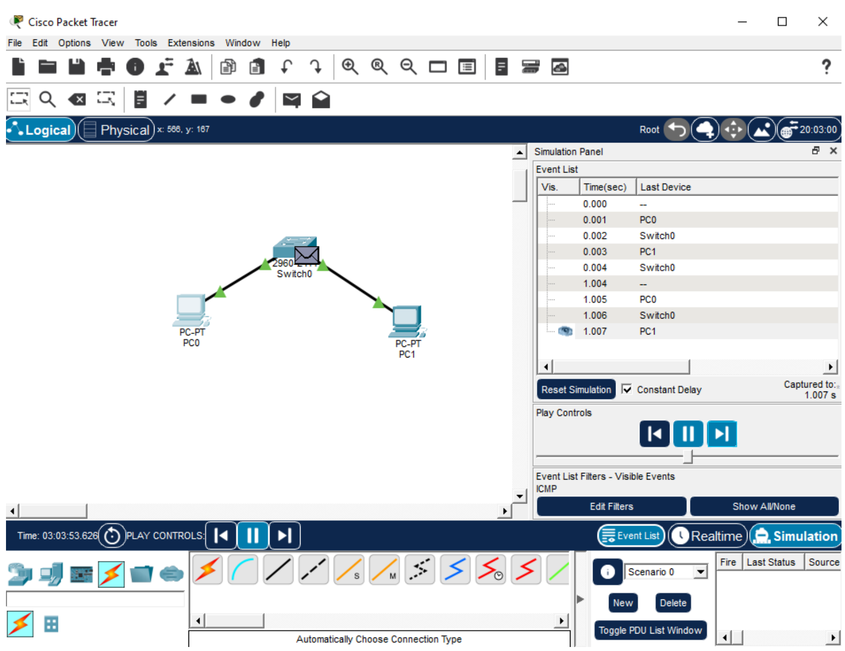
23. If the connection is working, you will see **Reply from 192.168.0.2**. This means PC0 successfully reached PC1.
24. If you see **Request timed out,** check that bot IPs are configured correctly and the cables are connected properly.

Note: ping uses ICMP (Internet Control Message Protocol) to send a small test packet from one device to another. It is the most basic and commonly used network troubleshooting tool. You will use this constantly in real IT and networking work.

Part 6 – Simulation Mode

Simulation Mode lets you visually see network traffic moving between devices in real time. It is a powerful learning tool for understanding how protocols actually work.

25. Click **Simulation** in the bottom right corner of Packet Tracer to switch to Simulation Mode. A panel will open on the right.
26. By default, all event types are selected which can get overwhelming. Click **Show All/None** to deselect everything.
27. Click **Edit Filters** and check **ICMP** only. Close the window.
28. Go to **PC0 > Desktop > Command Prompt** and run the ping command again: ***ping 192.169.0.2***
29. You will see animated envelope moving across the topology. These represent the ICMP packets being sent from PC0 to PC1.



30. Click on any envelope to see its details, including the **OSI model layer** it passed through and the **source and destination IP** in the PDU details.

Note: This visual representation is one of the best ways to understand networking concepts. When you get deeper into protocols like ARP, OSPF, or DHCP, Simulation Mode makes it much easier to understand what is happening behind the scenes.

Key Takeaways

- **Packet Tracer is free** and one of the best tools available for learning networking, especially for networking or CCNA preparation
- **Logical View vs. Physical View –** you will spend most of your time in Logical View, but Physical View is useful for visualizing real world network scale
- **Devices on the same network** can communicate directly without a router. Both PCs needs to be in the **192.169.0.2/24** range for this to work.
- **Ping is the most basic connectivity test –** if ping works, the basic network connection is functioning properly
- **ICMP** is the protocol behind ping. A small test packet sent from one device to another to check if it is reachable
- **Simulation Mode** is a powerful learning tool. Use it to visually understand, how traffic moves and how protocols behave across a network